

Disjoint Set Union

September 6, 2026

A (ID 2200110)

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Department of Computer Science and Engineering

Outline

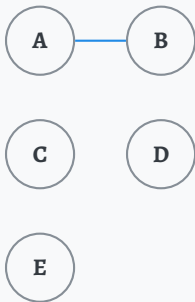
Introduction

A Familiar Problem



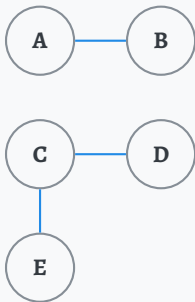
- Suppose there are five students in a classroom.
- At first, none of them know one another.

A Familiar Problem



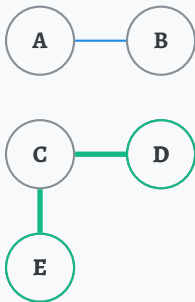
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- A and B become friends.

A Familiar Problem



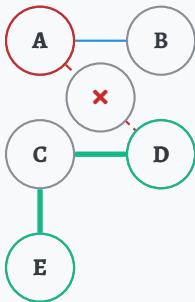
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- **Are D and E friends?** Yes — indirectly, through C, they belong to same friend circle.

A Familiar Problem



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- At first, none of them know one another.
- A and B become friends.
- Separately, C become friends with D and E.
- **Are D and E friends?** Yes — indirectly, through C, they belong to same friend circle.
- What about **A and D**? No — they belong to two different circles entirely.

Larger Networks

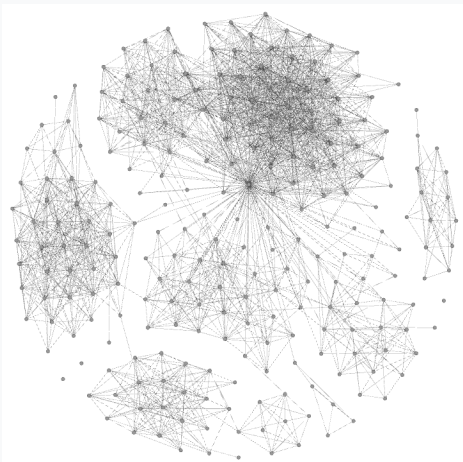


Figure: *Social Media Network Visualization^a*

- Consider a large social network, or equally, a large computer network, with countless members.

^aPinterest: Visualise Your Facebook Friend Network

Larger Networks

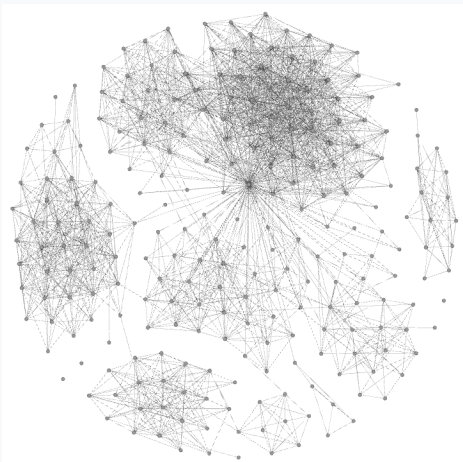


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- We need to answer **efficiently**: **are any two given members connected?**

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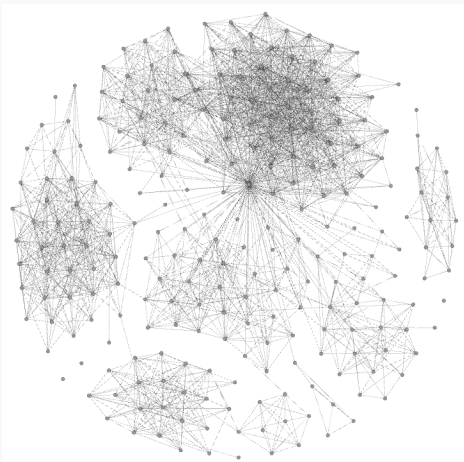


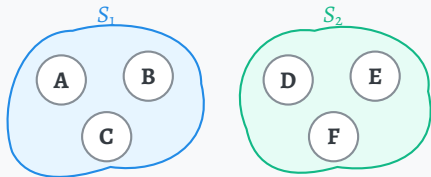
Figure: *Social Media Network Visualization*^a

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- We need to answer **efficiently**: **are any two given members connected?**

This is exactly what *Disjoint Set Union* answers.

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What Is a Disjoint Set Union?



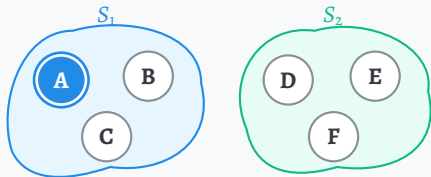
Definition

- Sets are **disjoint** if they share no elements:

$$S_i \cap S_j = \emptyset \quad \text{for } i \neq j.$$

- **Disjoint Set Union** or **DSU** are a collection of such disjoint groups.

What Is a Disjoint Set Union?



Here,

- A is the representative of S_1

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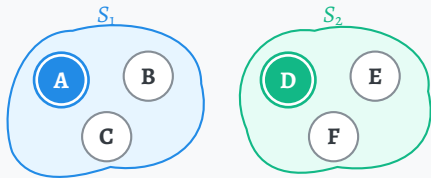
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- D is the representative of S_2

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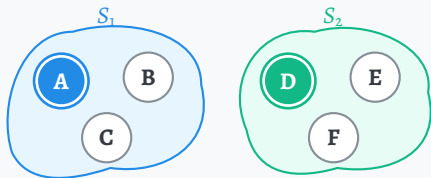
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Same representative $\iff$ same set.